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References

Sensory Processing Alterations as a Transdiagnostic Phenotype: Evidence from fMRI, Functional Connectivity, and Salience Networks

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Background and Aims

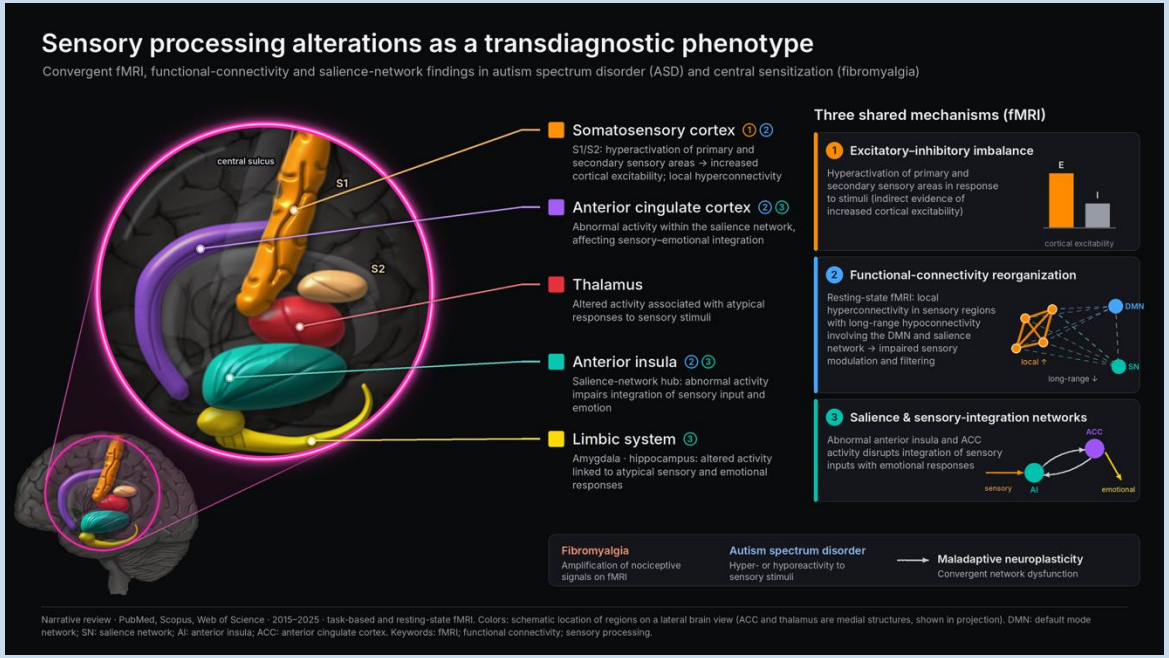
Alterations in sensory processing constitute a central feature across multiple neurological and neuropsychiatric conditions,including autism spectrum disorder(ASD)and central sensitization syndromes such as fibromyalgia. Recent evidence from functional neuroimaging (fMRI) has demonstrated that these conditions share convergent patterns of dysfunction in brain networks involved in sensory integration and salience attribution. Notably, alterations have been identified in regions such as the somatosensory cortex,insula,thalamus,and limbic system, which are associated with atypical responses to sensory stimuli. Furthermore, functional connectivity studies reveal large-scale reorganization of neural networks,suggesting a shared neurobiological substrate.

Objective

To investigate and synthesize the shared neurobiological mechanisms underlying conditions associated with sensory processing alterations, with emphasis on findings from functional neuroimaging(fMRI), brain connectivity, and neurotransmission.

Methods

Narrative review based on databases such as PubMed, Scopus, and Web of Science databases, including studies published between 2015 and 2025. Original studies, systematic reviews, and meta-analyses addressing sensory processing and neurobiological mechanisms were included, with a focus on studies employing fMRI (both task-based and resting-state). Search terms included “sensory processing,” “fMRI,” “functional connectivity,” “autism spectrum disorder,” and “central sensitization.”



Results

The analysis identified three principal shared mechanisms, supported by fMRI findings. First, an excitatory–inhibitory imbalance, indirectly evidenced by hyperactivation of primary and secondary sensory areas in response to stimuli, suggesting increased cortical excitability. Second, alterations in functional connectivity, with resting-state fMRI studies demonstrating local hyperconnectivity in sensory regions alongside hypoconnectivity across long-range networks, particularly involving the default mode network (DMN) and the salience network, impairing sensory modulation and filtering. Third, dysfunction within salience and sensory integration networks, with abnormal activity in the anterior insula and anterior cingulate cortex affecting the integration of sensory inputs and emotional responses. Additionally, in conditions such as fibromyalgia, amplification of nociceptive signals is observed on fMRI, whereas in ASD, patterns of hyper- or hyporeactivity to sensory stimuli are evident,both associated with maladaptive neuroplasticity.

Conclusions

The integration of clinical and functional neuroimaging findings indicates that different conditions characterized by sensory alterations share a convergent pattern of dysfunction across brain networks. fMRI plays a central role in identifying these biomarkers,contributing to a transdiagnostic understanding and supporting the development of therapeutic strategies aimed at modulating neural connectivity and excitability.